

Emmanuel Idoko

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EDUCATION

University of Lagos

Bachelor of Science in Computer Science; CGPA: 4.68/5.0

Lagos, Nigeria

Expected 2027

- **Relevant Coursework:** Discrete Mathematics, Data Structures, Computational Methods, Linear Algebra, Calculus, Python/C Programming, Net Centric Computing, Algorithm and Complexity, Compiler Construction, Concurrent Programming, Operations Research, Systems Analysis, Operating Systems, Machine Assembly Language.

African Computer Vision Summer School (ACVSS)

3rd ACVSS, Google AI Community Center

Accra, Ghana

Jul 2026

- **Sessions:** 10 lectures and 8 practical sessions covering geometry and mathematics for vision, deep learning and AI, and ethics in computer vision, alongside 8 keynotes, a hackathon, and a poster session.

Udacity

AI Programming with Python Nanodegree

Remote

Oct 2022 – Jan 2024

PUBLICATIONS & ONGOING PROJECTS

SharpXR: Structure-Aware Denoising for Pediatric Chest X-Rays

Accepted — MIRASOL Workshop, MICCAI 2025 | [arXiv:2508.08518](https://arxiv.org/abs/2508.08518)

- Contributed to the experimental evaluation of a structure-preserving denoising method for noisy pediatric chest X-rays, benchmarking seven baseline models (REDCNN, DnCNN, HFormer, ResUNet++, Attention U-Net, Sharp U-Net, BM3D).
- Demonstrated improved downstream pneumonia classification accuracy from 88.8% to 92.5% using SharpXR-denoised images.
- Wrote the abstract and introduction; performed proofreading and editing of the full manuscript.

Research Poster Presentation

Machine Intelligence Research Group International Conference (MIRG-ICAIR 2025)

- Presented a research poster on SharpXR at MIRG-ICAIR 2025.

Cross-Modality Attention Fusion for Chest X-ray Diagnosis

Ongoing

- **Objective:** Investigate whether fusing features from chest X-ray images and textual radiology reports improves diagnostic accuracy.
- **Methodology:** Designing a cross-attention transformer model to learn joint representations of visual and textual data, exploring its effectiveness in identifying subtle pathologies missed by image-only models.

VAMAE: Vessel-Aware Masked Autoencoders for OCT-Angiography

Accepted — 28th International Conference on Pattern Recognition (ICPR 2026) | [arXiv:2604.06583](https://arxiv.org/abs/2604.06583)

- **Objective:** Develop a self-supervised pre-training method for OCT-A image analysis that learns robust representations of retinal vasculature.
- **Methodology:** Proposing a masked autoencoder (MAE) variant with a vessel-aware loss function that prioritizes reconstruction of fine vascular structures to improve downstream segmentation and classification.

RESEARCH EXPERIENCE

AidCare

AI Research Lead & Co-Founder

Lagos, Nigeria (Hybrid)

May 2025 – Present

- Architected a dual-mode Retrieval-Augmented Generation (RAG) system to provide context-aware clinical decision support, leveraging distinct knowledge bases for physicians and community health workers.
- Engineered a semantic search pipeline using Sentence Transformers and FAISS to index and query a heterogeneous corpus of medical textbooks and Nigerian national guidelines.
- Developed a multi-stage LLM prompting framework (Google Gemini) for grounded reasoning, enabling the synthesis of patient data with retrieved knowledge to generate differential diagnoses.
- Designed a multi-modal data ingestion pipeline integrating local Whisper AI for transcription and Tesseract OCR for patient document processing.

ML Collective

Independent ML Researcher

Remote

Sep 2024 – Present

- Engaged in weekly critical analysis and presentation of seminal and contemporary research papers in machine learning, with a focus on medical imaging and clinical NLP.
- Collaborated within focus groups to identify open research questions, formulate novel hypotheses, and design experimental protocols for computer vision research projects.

Wema Bank

Data and AI Internship

Lagos, Nigeria (Hybrid)

May 2024 – Dec 2024

- Researched and built a governance-compliant pipeline to extract and analyze Play Store reviews, producing actionable insights for product teams.
- Contributed to the development of an AI-powered knowledge chatbot for Microsoft Teams, enabling 2,000+ staff to instantly access product and process information, reducing support query time.

ENGINEERING EXPERIENCE

HabariPay – GTCO

Software Engineer Internship

Lagos, Nigeria (Hybrid)

Apr 2025 – Jun 2025

- Built a full-stack vendor guide app (Next.js, Node.js, PostgreSQL) for a food festival, used by 260+ attendees and generating 2,100+ page views.
- Automated ingestion of vendor data from 2 sources into a centralized SQLite DB, processing hundreds of records with high consistency.
- Implemented a TF-IDF-based recommendation engine and Gemini-powered summaries, improving user click-through rate by 7%.

NexaScale

Cloud Engineer Internship

Remote

Feb 2023 – May 2023

- Automated deployment of 30+ AWS resources with Terraform, reducing manual setup by 75%.
- Developed a microservice-based application with Node.js, containerized with Docker, and deployed to production with Kubernetes (AKS).
- Streamlined deployments with a CI/CD pipeline (Azure DevOps, Git, Docker) for fast and reliable releases.

ZuriChat

Software Engineer & Lead Internship

Remote

Jul 2021 – Oct 2021

- Led a team to ship real-time music and chess rooms using WebRTC, boosting user engagement by 20%.
- Improved backend performance by 50% through caching, code refactoring, and API optimization.
- Managed multi-team deployments with GitHub Actions and Jenkins, ensuring stable integrations across features.

PROJECTS

Virtue Foundation Intelligence Platform

Python, LangGraph, FAISS, GPT-4o-mini, Next.js

- Built an end-to-end agentic AI platform to detect medical deserts and analyze healthcare gaps across 797 Ghana facilities in all 16 regions.
- Designed an Intelligent Document Processing (IDP) pipeline using GPT-4o-mini to extract structured capability data from unstructured facility records, with per-field confidence scoring (0-99%).
- Architected a LangGraph supervisor multi-agent system classifying natural language queries into 6 categories and routing to specialized sub-agents backed by a FAISS vector store on OpenAI embeddings.
- Identified 10 medical deserts across 16 regions, flagged 43 data anomalies across 39 facilities, and generated AI-powered deployment recommendations; integrated ElevenLabs voice I/O for field accessibility.
- Won 2nd place at Databricks x Hack-Nation Global AI Hackathon.

Kinetix

Python, FastAPI, Next.js, TypeScript, Gemini API, WebSockets

- Built a real-time AI personal trainer using Gemini's vision capabilities to analyze webcam exercise footage frame-by-frame, classifying posture and joint alignment as safe, suboptimal, or unsafe.
- Designed a dual-persona prompt architecture: an internal Analyst agent performs biomechanical reasoning while an external Coach agent delivers concise real-time verbal coaching via text-to-speech.
- Implemented WebSocket communication between FastAPI backend and Next.js frontend for low-latency feedback, with skeleton overlay and color-coded form indicators (green/yellow/red).
- Generates post-session summaries with rep counts, recurring problem areas, and corrective exercise recommendations.

PheraCAM

Python, React.js, Azure, OpenAI API

- Developed a real-time face recognition system leveraging OpenCV and Azure cloud services for scalable, low-latency processing.
- Achieved >95% recognition accuracy with <150ms response time.
- Won 2nd place among 50+ teams at Hacklab's Hackathon, recognized for technical innovation and robustness.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript/TypeScript, SQL, C, Bash
- **ML & Imaging:** PyTorch, Hugging Face (Transformers, Sentence Transformers), Scikit-learn, FAISS, Pandas, NumPy
- **Frameworks/Tools:** FastAPI, Flask, Node.js, Next.js, Docker, PostgreSQL, MongoDB, DigitalOcean, Git, Linux

AWARDS & COMMUNITY LEADERSHIP

Project Lead | Unilag Data Community

Dec 2024 – Present

- Developed and delivered a comprehensive workshop series (Python, SQL, Power BI, Excel), successfully training over 250 students.
- Mentored beginner and intermediate members on project scoping, methodology, and technical implementation.

Deputy Trainings & Competitions Lead | GDSC UNILAG

- Coordinated internal hackathons and competitions, fostering innovation and practical problem-solving.
- Designed and led technical training sessions and hands-on workshops.

SCHOLARSHIPS

- Jim Ovia Undergraduate Scholarship (2025–2027): National scholarship recognizing top performance in STEM.
- Squad Hackademy Undergraduate Scholarship (2024–2027): Recognizing top performance in programming.
- PICFI Undergraduate Scholarship (2025–2027): National scholarship recognizing top performance in STEM.